PERSONAL REFLECTION

As Artificial Intelligence Advances, What Will Become of the Next Generation?

A Personal Reflection on the Article *AI Disruption and Falling Costs in Theory Research*

Shlomi Hazan

Introduction

Personally, I can say that I found the article fascinating. The insights it presented, the research methodologies, the discussion of problems and the solutions that artificial intelligence may offer, and above all, the very exploration of a subject so significant to the future of human research, sparked a great deal of interest in me. However, alongside my admiration for the technological progress described in the article, I found myself reflecting on questions concerning not only the future of research, but also **my own future and that of many young people** seeking to enter the technological world.

The Changing Role of Researchers

One of the central insights I took away from reading the article concerns the potential transformation of the roles of researchers and individuals with advanced academic qualifications. What will become of the roles of master's degree holders, PhD graduates, and professors in a world where an increasing share of research work may be carried out by artificial intelligence? Will research, development, and academic writing remain at the heart of their work, or will the focus of their responsibilities shift?

The findings presented in the article point to significant progress in the ability of artificial intelligence systems to address complex research problems at a relatively low cost. This raises, in my view, a troubling possibility: in the future, a significant part of researchers' responsibilities may shift from **independently creating new knowledge to examining, validating, and critically evaluating** research outputs produced by artificial intelligence.

As the discussion in the article suggests, the problem is not necessarily the ability to produce research outputs quickly and in large quantities, but also the need to ensure that these outputs are accurate, reliable, and scientifically valuable. As the pace of producing such outputs increases, so too may the need for knowledgeable and experienced researchers capable of critically evaluating them.

To me, this represents a fundamental and concerning change. What happens to a researcher's professional identity when part of the work that once defined their profession is now performed by an artificial system? Are we approaching a transition from a researcher whose role is to create, innovate, and discover, to one whose primary responsibility is to understand, examine, validate, and approve research they did not conduct themselves, but that was instead produced by artificial intelligence?

I do not mean to suggest that critically evaluating and validating research are less valuable responsibilities. On the contrary, their importance may only continue to grow. Nevertheless, I find it difficult to ignore the question of what will happen to the place of human creativity, innovation, and curiosity in the research process if an increasing share of it is carried out by artificial intelligence.

The Younger Generation: Entry-Level Developers

The rapid leap in the capabilities of artificial intelligence systems, as reflected in the article, may raise concerns among researchers and highly educated professionals. Yet it was precisely at this point that I found my thoughts moving in a different direction: not toward established researchers, but toward the younger generation, entry-level developers, and those who have yet to take their first steps into the software and computer science industries.

I am currently approaching the end of my bachelor's degree in Computer Science and the beginning of my master's degree, and I admit that I am concerned about my professional future in this field.

As things appear today, artificial intelligence is capable of performing a considerable portion of the tasks that were previously assigned to entry-level software developers. Writing basic code, software testing, identifying and fixing errors, and even routine maintenance tasks are just a few examples.

These are precisely the tasks through which young developers traditionally gained experience, learned to navigate their working environments, and gradually progressed toward more complex responsibilities. If artificial intelligence systems can perform a significant portion of these tasks, the question arises: what place will an entry-level developer have in the future job market?

A Rising Barrier to Entry

However, my concern is not limited to the possibility that artificial intelligence will reduce the demand for entry-level positions. It becomes broader when I return to the conclusions and questions raised by the article regarding the world of research. If even experienced researchers, with advanced academic qualifications and years of experience, may face fundamental changes in their roles as a result of artificial intelligence, what does this mean for someone who holds only a bachelor's degree?

Are we approaching a world in which a bachelor's degree in Computer Science will no longer be sufficient to enter the industry? Will the level of knowledge and skills required of candidates continue to rise, until we are expected to possess a level of expertise that was once primarily associated with holders of advanced degrees?

Even today, studies in technological and engineering fields are considered challenging academic paths, requiring analytical abilities, mathematical thinking, perseverance, and the capacity to deal with complex problems. In my opinion, technological development and early exposure to technology have contributed to making skills that were once considered complex more intuitive for younger generations. Children today grow up in a digital environment, encounter technological systems from an early age, and develop a more natural familiarity with the tools surrounding them. At the same time, the changing demands of the labor market encourage many people to acquire more advanced education and skills.

Yet even if we assume that human capabilities and access to knowledge are improving, one significant question remains in my mind:

> **Can the pace of human development, in terms of acquiring knowledge and skills, keep up with the pace of artificial intelligence development? And will the qualifications that have so far been considered sufficient for entering technological professions continue to be enough in the future?**

Expertise and Critical Evaluation as Core Skills

If we follow this line of thought, it is possible that the software industry will increasingly demand individuals with deep expertise, research capabilities, and the ability to critically evaluate outputs generated by artificial intelligence systems.

To evaluate a complex research or technological output, it is not enough to know how to operate the system that produced it. One must understand the problem it was attempting to solve, recognize the assumptions on which it is based, know how to read and evaluate research, examine proofs, identify logical flaws, refute or substantiate claims, and above all, **know how to challenge the conclusions reached by artificial intelligence**.

These are skills that require significant knowledge, experience, and depth of understanding. Some are acquired during a bachelor's degree, while others develop later through advanced studies, professional experience, and research work. In my estimation, as artificial intelligence systems become more complex and autonomous, the demand for these skills may increase. As a result, advanced education and research experience may become an even more significant advantage in the job market, and perhaps even a prerequisite for certain positions.

Will Fewer People Choose These Fields?

And this brings me to the question that troubles me most:

> **Could Software Engineering and Computer Science become fields that fewer and fewer people want to study, precisely because the barriers to entry will become too high?**

Imagine a young person standing at the point of choosing their academic path and professional future. In the past, studying Computer Science was perceived by many as choosing a challenging yet promising path: investing several years in education, acquiring an in-demand profession, and having the opportunity to enter an industry offering rewarding employment conditions and salaries. But what happens if that same person knows in advance that a bachelor's degree may not be enough? That they may need to invest additional years in education, acquire much deeper expertise, and compete increasingly with artificial intelligence systems, just to have a chance of entering an entry-level position?

Would they still choose that path?

As a Computer Science student, I am already exposed to these concerns in my immediate surroundings. In conversations with lecturers, professors, and academic officials at my institution, I have heard about a decline in the number of students enrolling in Computer Science programs compared to previous years. I do not claim that this necessarily reflects a global trend, but in my view, the very existence of this discussion indicates a certain shift in how young people perceive the future of the profession.

In the past, a degree in Computer Science was almost perceived as a promising entry ticket into the high-tech industry. For many, it was a desirable, prestigious path with a relatively clear employment outlook. Today, at least based on the conversations I have and the concerns I hear around me, that feeling is no longer the same. Some fear that artificial intelligence will reduce the number of available jobs. Others worry that it will undermine the very elements that made the profession attractive in the first place: creativity, imagination, problem-solving, and intellectual challenge. And there are also those who simply wonder whether the long and significant investment in education will still be worthwhile.

These concerns do not seem imaginary or disconnected from reality to me. Even if we do not yet know what the job market will look like in a decade, the uncertainty is already being felt among students and young developers.

Implications for the Next Generation

And if this is what things look like today, what will happen in the future? What will happen to enrollment rates in Computer Science and Software Engineering programs when the level of knowledge, expertise, and achievement required to enter the industry continues to rise? What will happen to young people's motivation to choose a long and demanding professional path when, even at the end of it, an employment opportunity is not necessarily waiting for them? And above all, **what will happen to creativity, boldness, and the human desire to research and develop** if the next generation begins to believe that it no longer has a meaningful place in the process?

I believe this question is no less important than the question of how the work of established researchers will change. After all, if in the future we need people with deep expertise to critically evaluate, validate, and challenge artificial intelligence systems, but at the same time we reduce the opportunities for entry-level developers and researchers to gain experience and grow, how will we train the next generation of those very experts?

Perhaps the real challenge is not only which roles artificial intelligence will be able to replace, but also how we can ensure that the next generation will continue to learn, develop, and become experts in a world where some of the earliest stages of professional development are already being carried out by machines.

I am not claiming that Computer Science or Software Engineering are about to disappear. It is entirely possible that artificial intelligence will create new opportunities, new professions, and research possibilities that we could not have imagined before. But precisely because the future remains uncertain, I believe it is important to ask not only what technology will enable us to do, but also **who will be able to participate in the world it creates**.

After all, the fear of struggling to find employment is no longer an imaginary scenario for those at the beginning of their careers. One only needs to speak with students and young developers to understand how present this uncertainty is in their lives. So what will happen when we tell the next generation, right at the beginning of their journey:

> **You will have to study more, know more, specialize more, and invest much, much more, just to have a chance of finding your first job?**

Who would even want to enroll in such a degree program?

Concluding Question

If I had to summarize all the thoughts, concerns, and insights that emerged from reading this article in a single question, I would phrase it as follows:

> **If artificial intelligence raises the level of knowledge and skills required to enter technological professions, could we reach a point where fewer people choose to study these fields in the first place — precisely at a time when technology is advancing at an unprecedented pace?**

This reflection expresses my personal opinions, thoughts, and insights developed through reading the article, alongside my personal experience as a Computer Science student at the beginning of my professional and research journey. The arguments and predictions presented here reflect my own interpretation and do not necessarily represent the conclusions of the article's authors.